Model accuracy analysis

Portable metric summary. Open model_analysis.html for all SVG charts and full details.

model_name: majority_baseline
split: test
split_role: test
n: 59
classification.accuracy: 0.372881
classification.balanced_accuracy: 0.25
classification.macro_precision: 0.0932203
classification.macro_recall: 0.25
classification.macro_f1: 0.135802
classification.multiclass_brier: 1.25424

model_name: majority_baseline
split: validation
split_role: tuning
n: 57
classification.accuracy: 0.45614
classification.balanced_accuracy: 0.25
classification.macro_precision: 0.114035
classification.macro_recall: 0.25
classification.macro_f1: 0.156627
classification.multiclass_brier: 1.08772

model_name: phase_logistic
split: test
split_role: test
n: 59
classification.accuracy: 0.559322
classification.balanced_accuracy: 0.481785
classification.macro_precision: 0.494222
classification.macro_recall: 0.481785
classification.macro_f1: 0.475237
classification.multiclass_brier: 0.554114

model_name: phase_logistic
split: validation
split_role: tuning

n: 57

classification.accuracy: 0.596491

classification.balanced_accuracy: 0.538462

classification.macro_precision: 0.569048

classification.macro_recall: 0.538462

classification.macro_f1: 0.536463

classification.multiclass_brier: 0.508782

Retrospective descriptive research decision support; no clinical validation or guaranteed future accuracy.

Tuning data support model selection; calibration data fit transforms/intervals; only test data provide the untouched held-out estimate. Test inspection can influence subsequent experiments.

Metrics are recomputed from every stored prediction. Saved aggregate metrics are parsed and hashed but are not treated as evidence of accuracy.

Current registry snapshots may contain post-start updates and leakage. Completed-only and convenience samples can create selection bias; chronological splits do not establish prospective validity.

No regression accuracy percentage is defined. Duration errors use days. Binary decisions use probability >= 0.5; confusion matrices have actual rows and predicted columns.

Undefined values are null, displayed as unavailable. One-class splits do not identify class discrimination. Small held-out samples and imbalance can make estimates unstable; no uncertainty interval is invented.

Reliability uses ten equal-width bins. Raw and post-hoc calibrated probabilities are analyzed separately; calibration-cohort diagnostics can be optimistic.

Duration-by-phase subgroup metrics cannot be recomputed because saved duration prediction rows do not contain verified phase labels; stored subgroup aggregates are not substituted.

Charts show all stored points (overlapping points can obscure sample counts); metrics and bin counts include every row.

This task predicts recorded phase from protocol characteristics; it does not predict progression or trial success.

Combined, NA and unknown phases are excluded. Class support can be small or absent, and classes absent from training have zero predicted probability.

All held-out records count, including labels absent from training. Confusion matrices use the five canonical labels (rows actual, columns predicted); macro-F1 averages the union of actual/predicted labels; balanced accuracy averages recall for actual supported classes.

Validation scores are optimistic after hyperparameter selection; untouched test scores are the held-out estimate. No performance improvement is guaranteed.

Redaction removes explicit phase mentions but cannot eliminate indirect phase cues or post-start updates in current registry snapshots.

Temporal ordering does not establish that current registry labels or protocol features existed at trial start; baseline provenance must be established separately.

Educational registry-phase consistency check only; not clinical advice or regulatory phase assignment. Class probabilities are uncalibrated model outputs, not clinical success probabilities. Minimum sample guards do not establish scientific validity. RETROSPECTIVE SNAPSHOT: baseline feature and label availability is unverified; registry updates and indirect narrative cues may cause leakage. Do not claim prospective validation.